Stuart Mitchell

Curriculum Vitae

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Qualifications

Jan 2015 – Dec 2017 **Master of Philosophy, Mathematics**, *Australian National University*
Thesis: *A Graphical Calculus for Shifted Symmetric Functions*

Feb 2010 – Dec 2010 **Graduate Diploma in Education in Secondary Teaching**, *University of Canberra*

Feb 2005 – Jul 2009 **Bachelor of Science with First Class Honours (Mathematics)**, *Australian National University*

Research Interests

My research is in categorification and representation theory. I am particularly interested in Heisenberg categories and their generalisations, the diagrammatic calculi used to study them, the representation theory of degenerate affine Hecke algebras, and connections with shifted symmetric functions, free probability and asymptotic representation theory.

Current Position

31 Aug 2026 – Present **PhD Student, Mathematics**, *Australian National University*, Supervisor: Prof. Anthony M. Licata
I am a PhD student in the Mathematical Sciences Institute at the Australian National University, working in categorification and representation theory. My current work builds on my earlier paper with Henry Kvinge and Anthony Licata, which connected Khovanov's Heisenberg category with free probability and shifted symmetric functions.

Publications

2019 H. Kvinge, A. M. Licata and **S. Mitchell**, Khovanov's Heisenberg category, moments in free probability, and shifted symmetric functions, *Algebraic Combinatorics* **2**(1) (2019), 49–74. doi:10.5802/alco.32, arXiv:1610.04571

2017 H. Kvinge, A. M. Licata and **S. Mitchell**, Khovanov's Heisenberg category, moments in free probability, and shifted symmetric functions, *Séminaire Lotharingien de Combinatoire* **78B** (2017), Art. 63, 12 pp. (FPSAC 2017 extended abstract)

Employment History

23 Jul 2012 – 11 Dec 2026 **Mathematics Teacher**, *Radford College*
Full-time, permanent member of the mathematics faculty, teaching all levels of mathematics in Years 7–12, RaVE to Years 7–10, and Theory of Knowledge for the IBDP. Leadership roles included Head of House, acting Assistant Head of Mathematics, Theory of Knowledge Coordinator and understudy Relief Coordinator.

27 Jan 2011 – 20 Jul 2012 **Mathematics Teacher**, *The Canberra College*
Full-time, permanent member of the mathematics faculty, teaching a wide range of BSSS courses and running supplementary tutorials for International Baccalaureate students.